

Diffusion models

Carlos Cotrini and Juan Tarazona
Building AI/ML applications
2026



Four main tasks in computer vision

Regression and
classification of images



Cat? Yes

Dog? Yes

Bat? No

Feature extraction



Cat color: black

Dog color: brown

Cat's eye color:
green

Dog's eye color:
brown

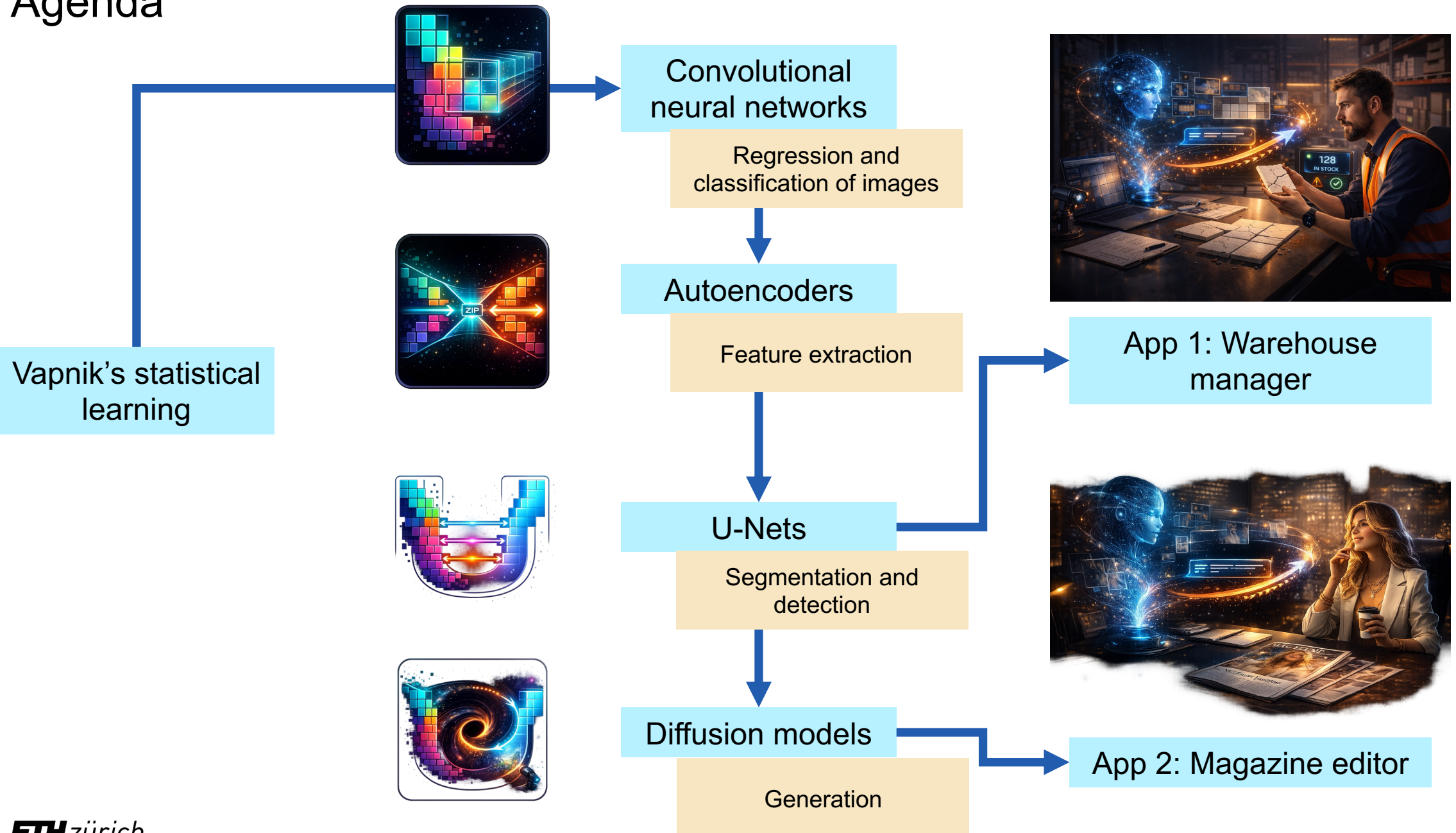
Segmentation and
detection



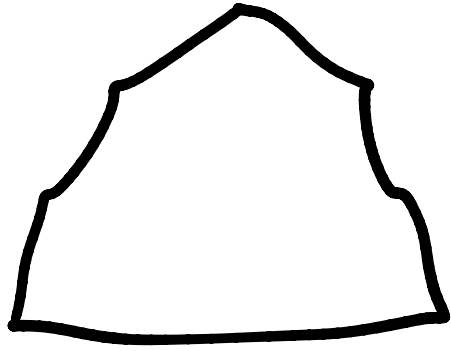
Generation



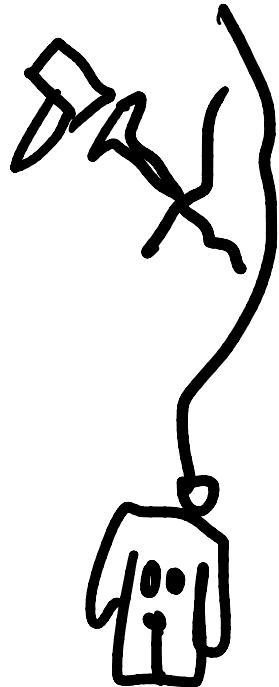
Agenda



Which are real? Which are fake?



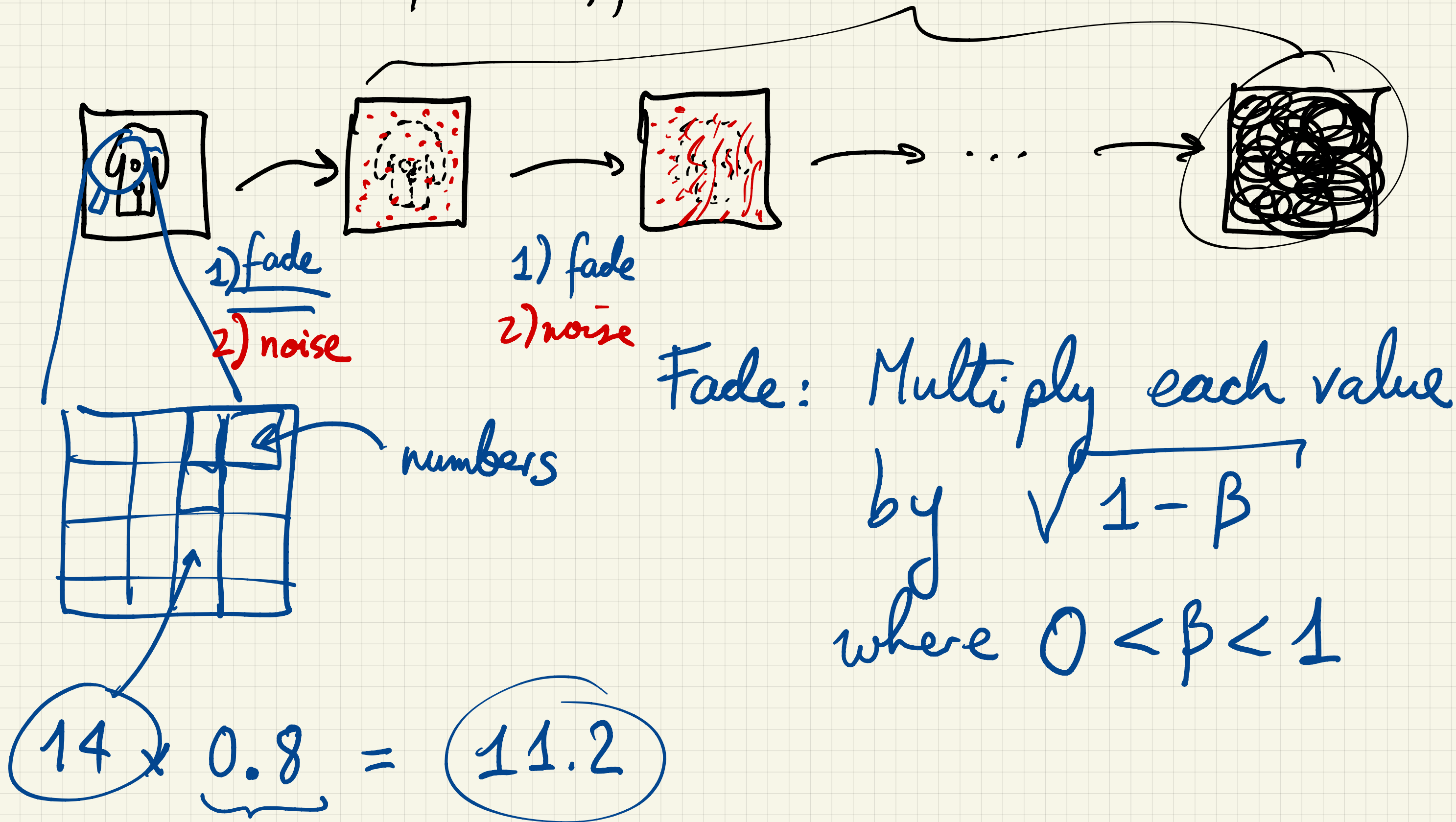
"Marble"



ETH zürich

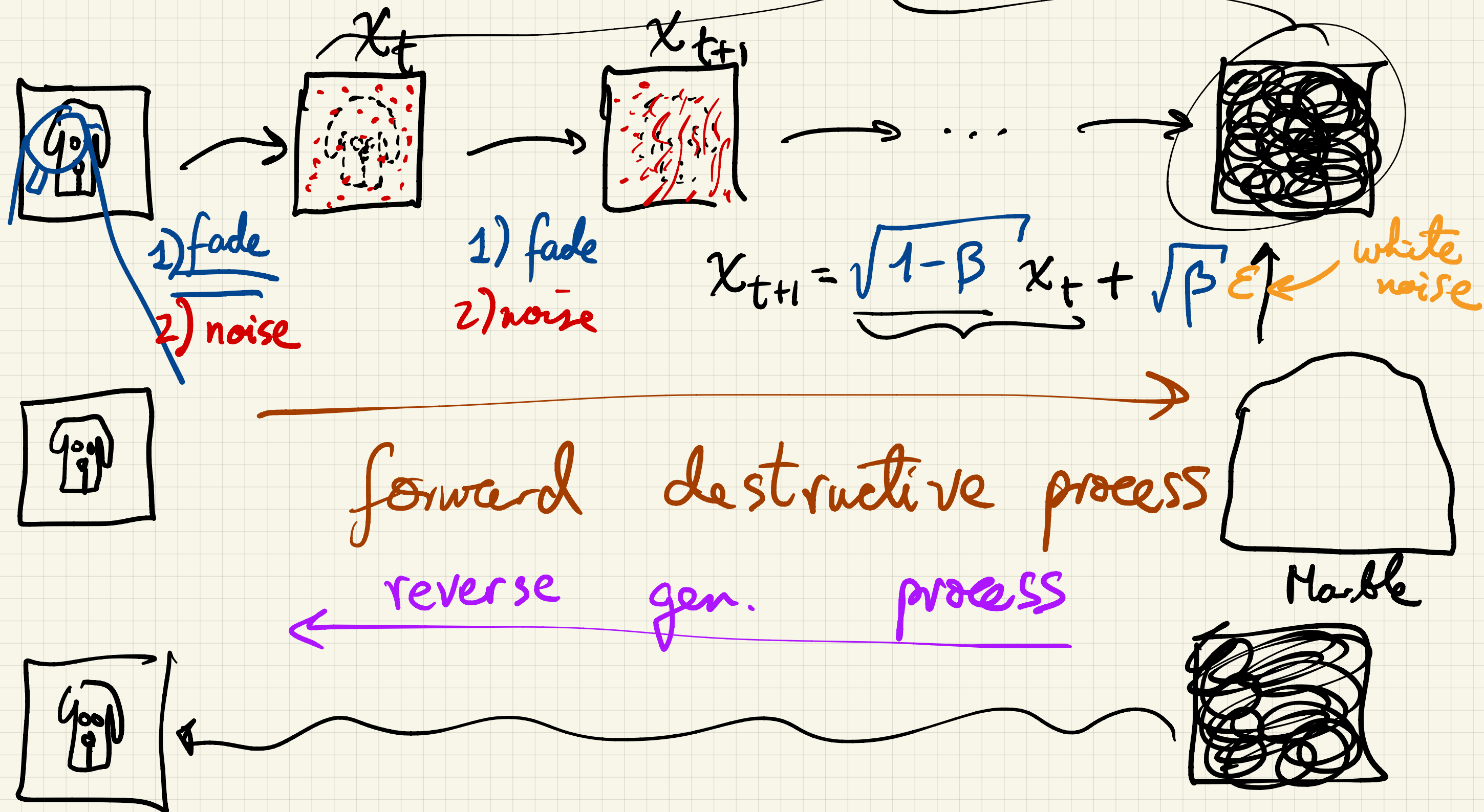


Essence of diffusion 1000 times

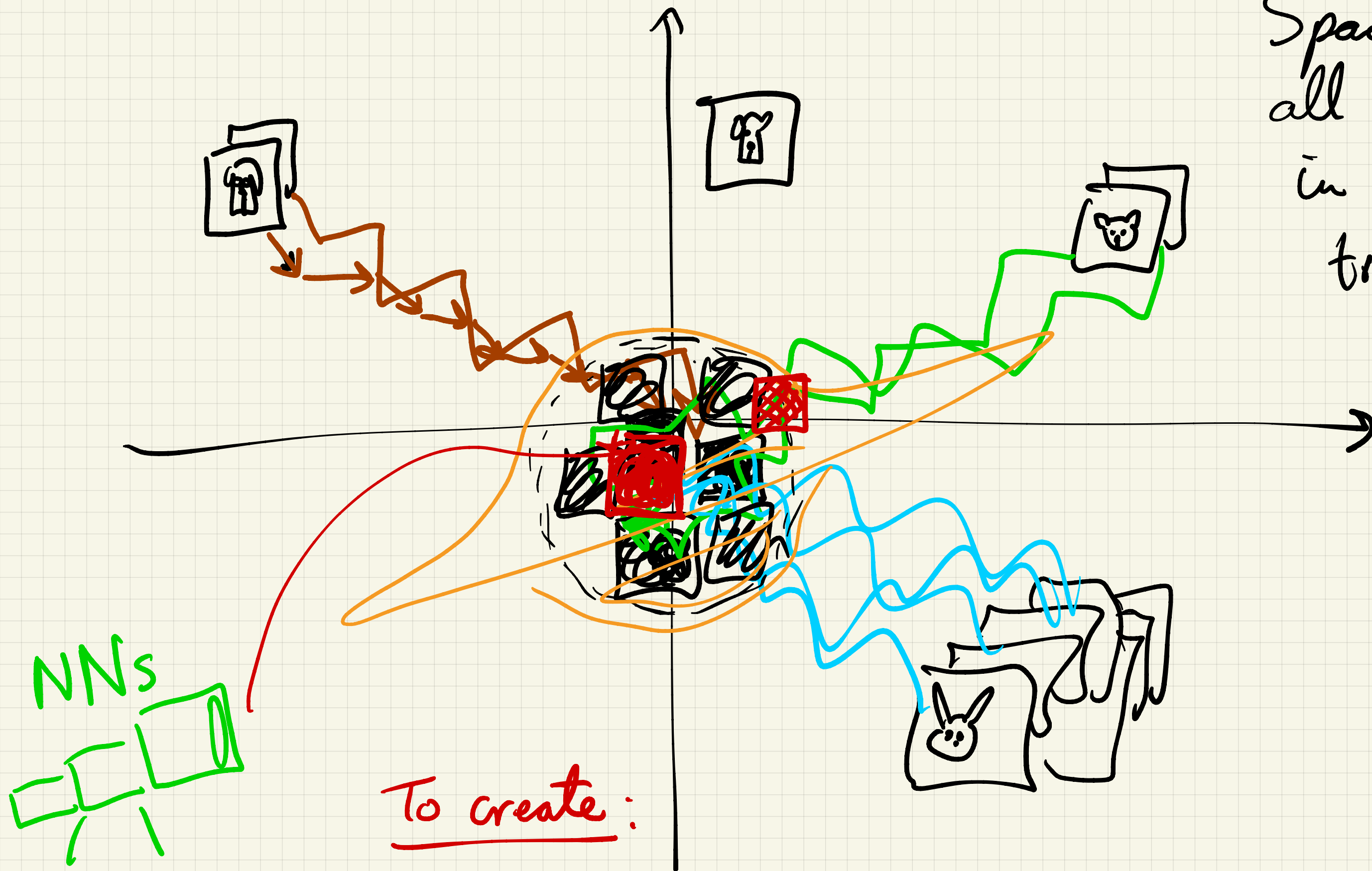


Essence of diffusion

1000 times

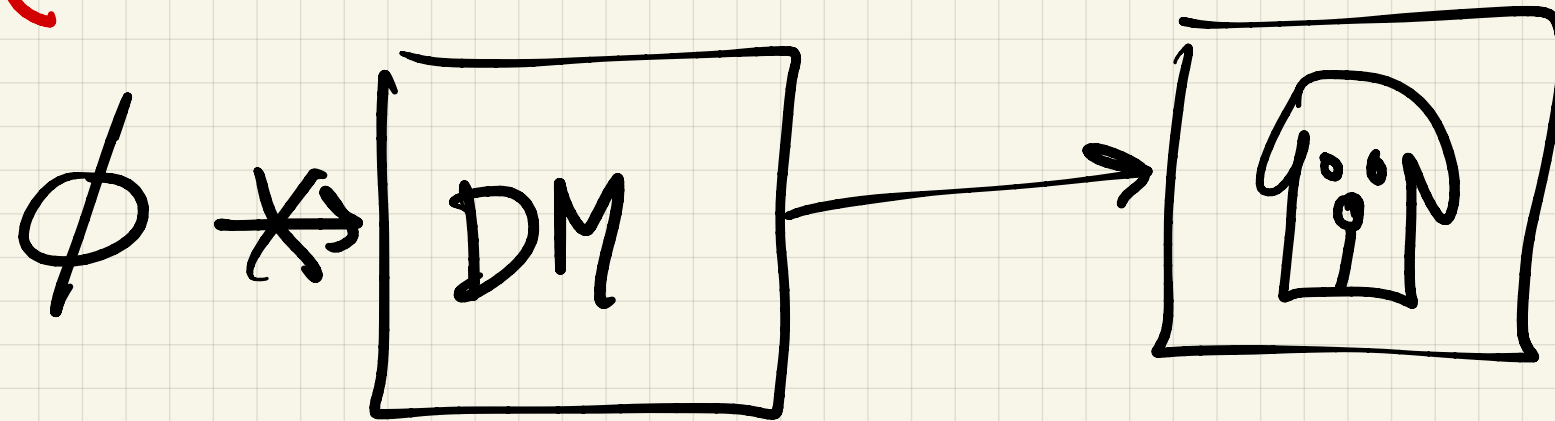


Space of
all imgs
in my
tr. dataset

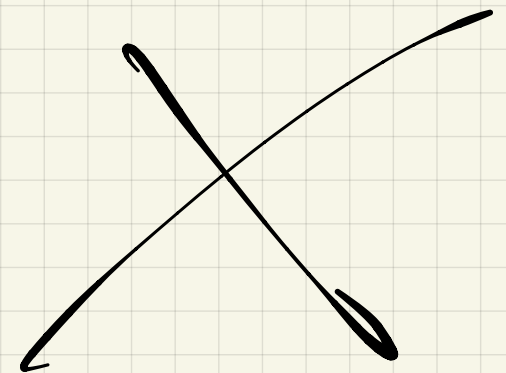
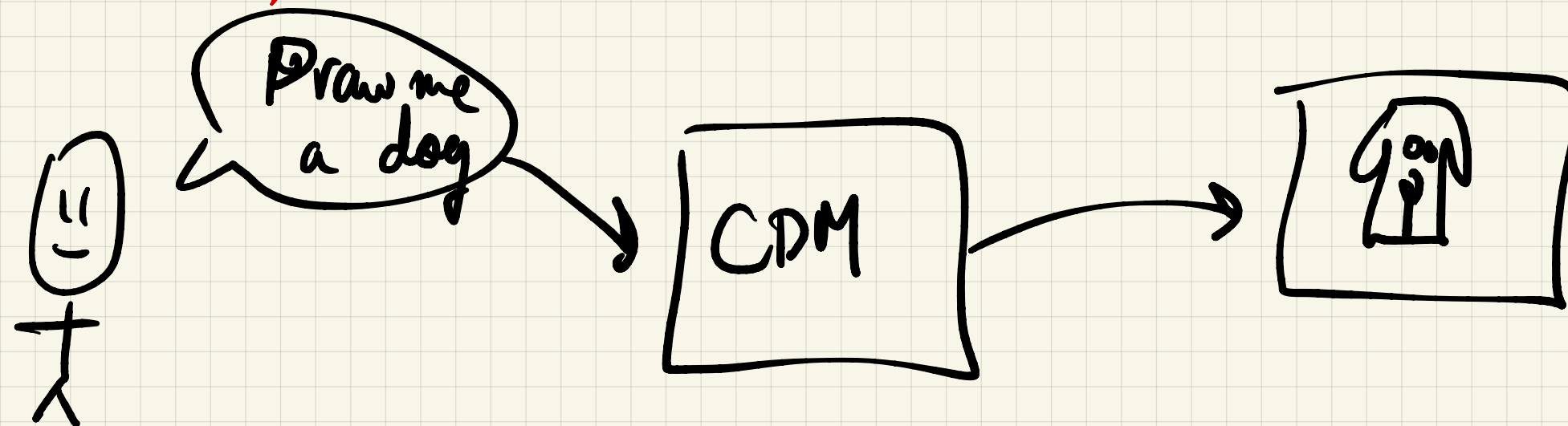


Important distinction

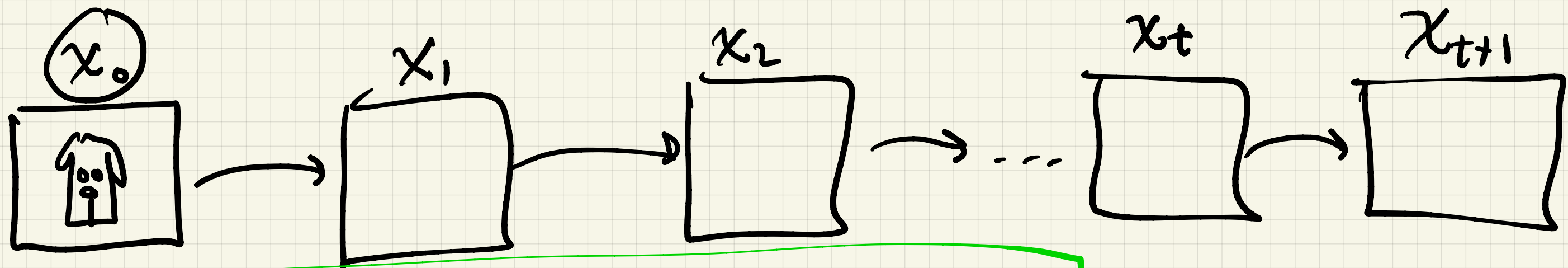
Diff model



Concl. diff. model



The forward destr. process



$$x_{t+1} = \underbrace{\sqrt{1-\beta_t}}_{\text{fading}} x_t + \underbrace{\sqrt{\beta_t}}_{\text{noise}} \epsilon_t$$

is between 0 and 1

$0 < \beta_t < 1$

ϵ_t is pure white noise.

One step forward

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t} \epsilon_t$$

$$\alpha_t = 1 - \beta_t$$

$$\bar{\alpha}_t = \alpha_0 \alpha_1 \dots \alpha_t = \prod_{i \leq t} \alpha_i$$

ϵ_t is pure white noise

Multi-step forward

The reverse gen. process

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon_t$$

$\alpha_t = 1 - \beta_t$
 $\bar{\alpha}_t = \alpha_0 \alpha_1 \dots \alpha_t = \prod_{i=0}^t \alpha_i$

ϵ_t is pure white noise

Multi-step forward

We can estimate how to reverse it

One step backwards

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_t \right) + \sqrt{\beta_t} z$$

z is pure white noise

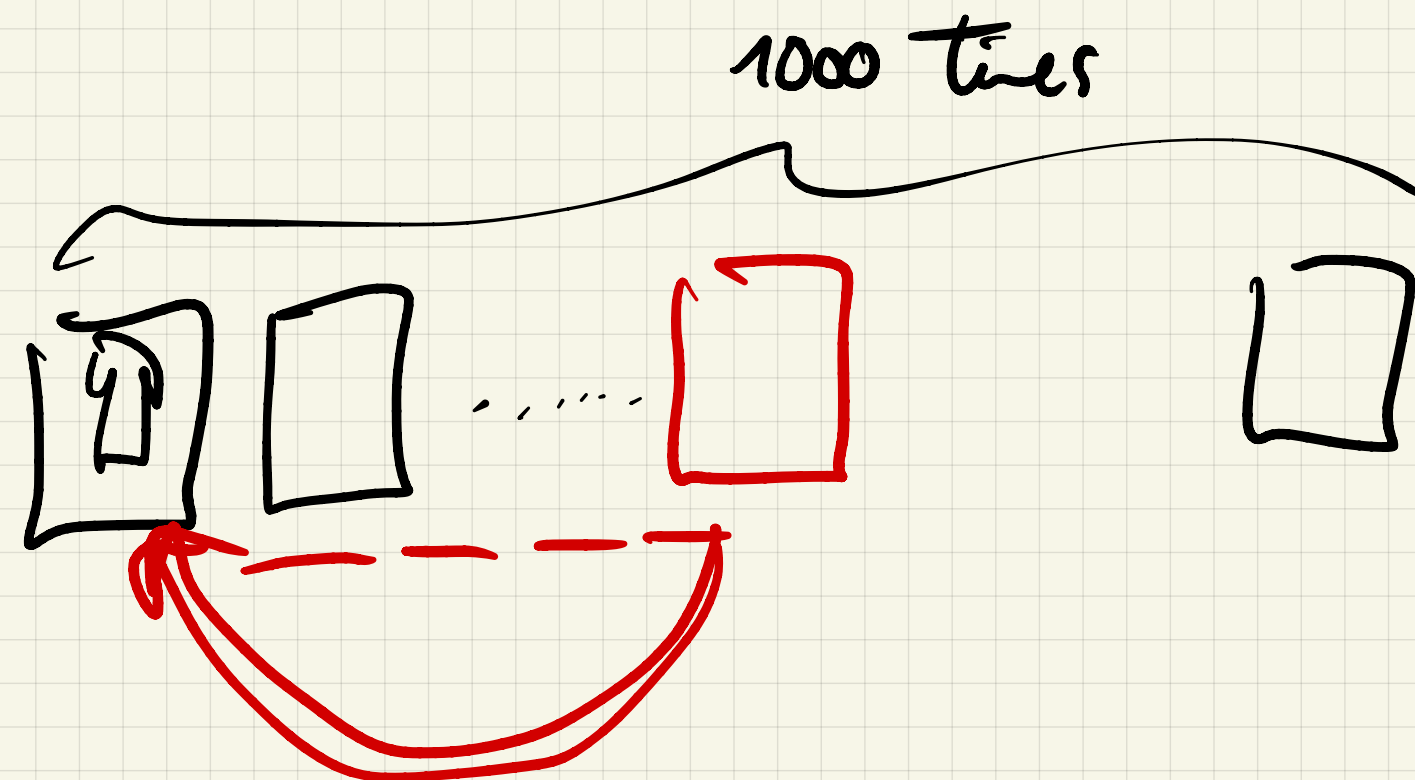
How to produce an image out of thin air...

1) Produce an image x_T with white noise

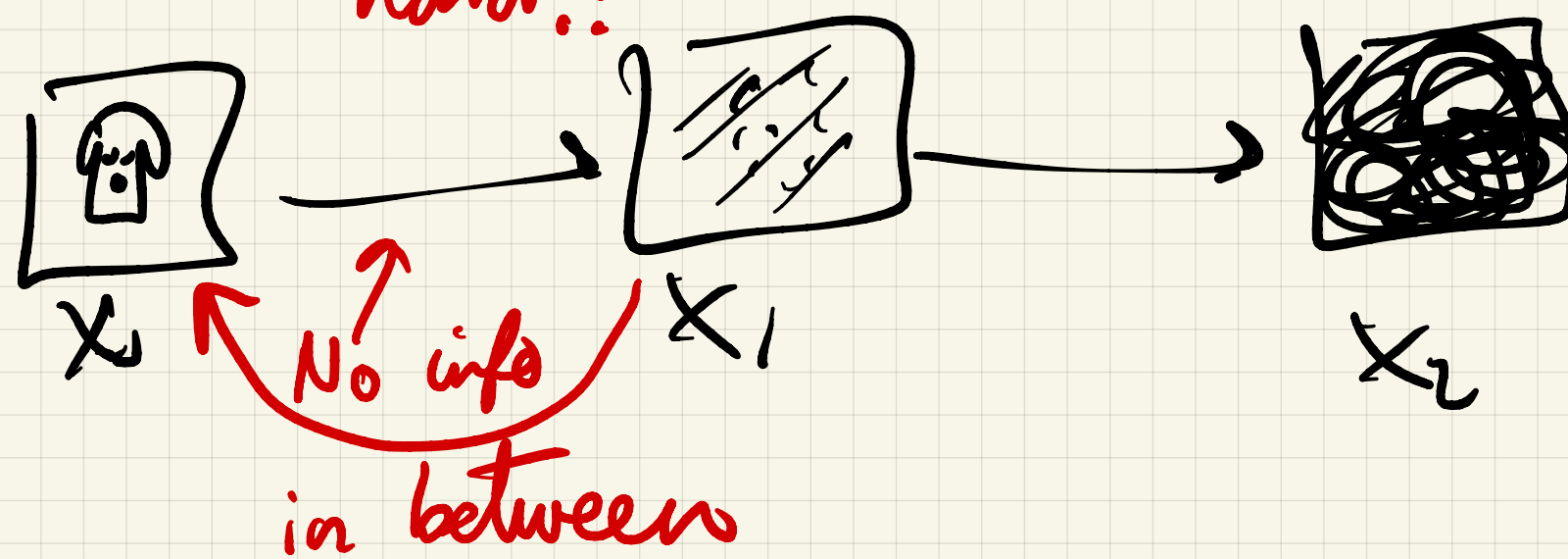
$$x_T = \text{[scribble]}$$

2) for $t = T, T-1, T-2, T-3, \dots, 3, 2, 1, 0$:

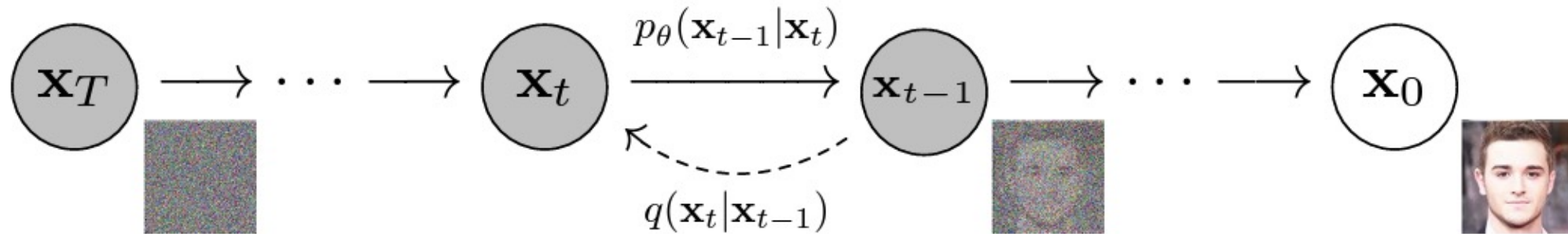
$$x_t = \frac{1}{\sqrt{\alpha_t}} \left(\frac{x_t}{\sqrt{1 - \bar{\alpha}_t}} - \epsilon_t \right) + \sqrt{\beta_t} z$$

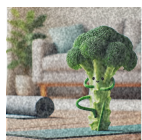
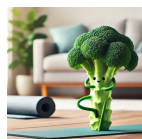


Alternative proposal
hard!!

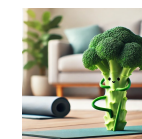
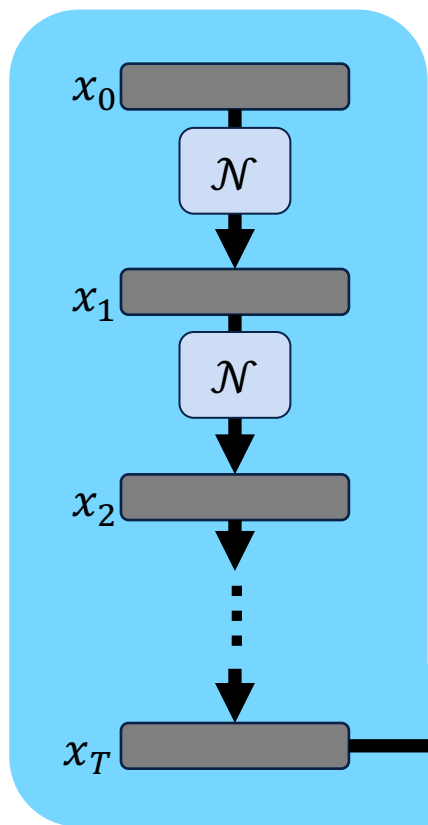


Denoising diffusion probabilistic models (DDPM) (Ho et al, 2020)

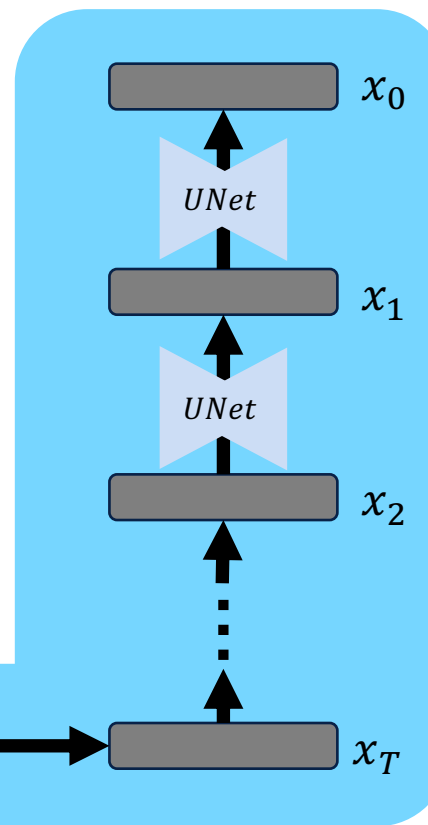




⋮



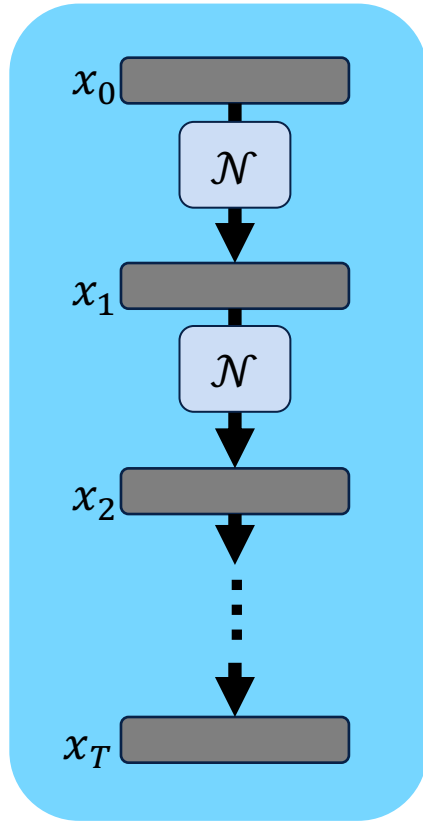
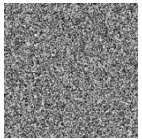
⋮



Forward process



⋮



For $t \leq T$, choose $\beta_t > 0$, let $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s \leq t} \alpha_s$. Then,

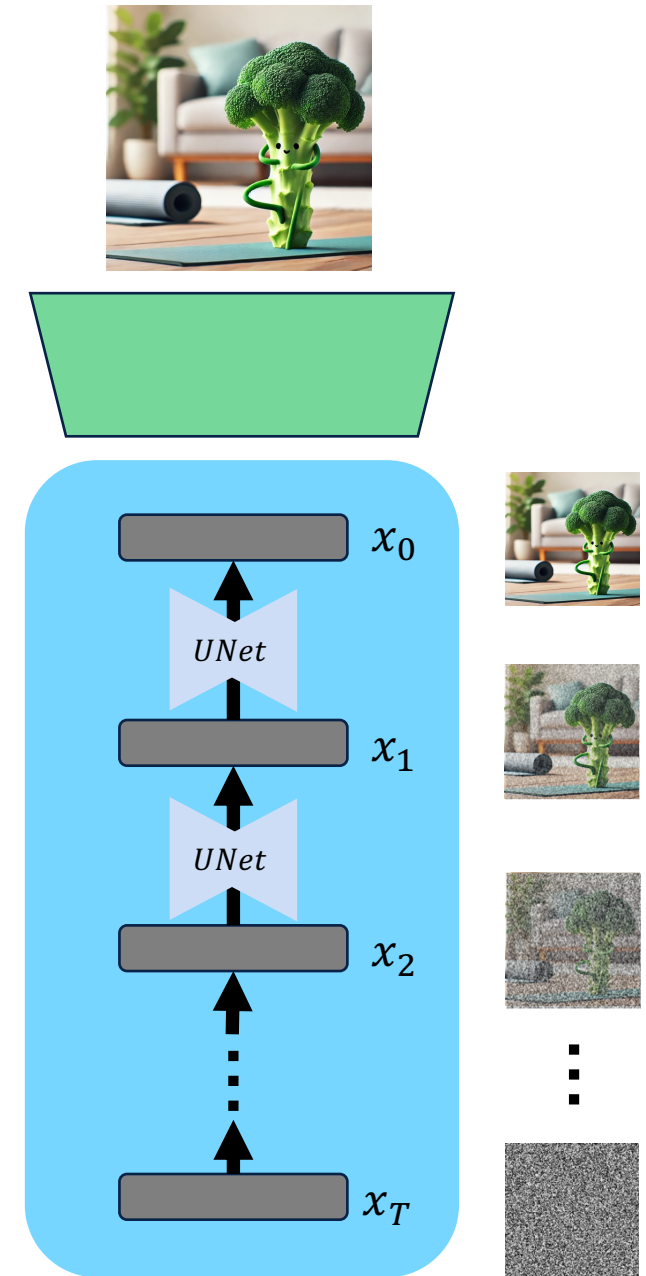
$$q(x_t | x_{t-1}) = \mathcal{N}(x_t | \sqrt{1 - \beta_t} x_{t-1}, \beta_t I)$$
$$q(x_t | x_0) = \mathcal{N}(x_t | \sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t) I)$$

Reverse process

- For each $t \leq T$, the UNet tries to reconstruct **all noise** $\epsilon_\theta(x_t, t)$ that was added to form x_t from x_0 .
- We then try to reconstruct x_{t-1} as follows

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1-\alpha_t}{\sqrt{1-\alpha_t}} \epsilon_\theta(x_t, t) \right) + \sigma_t z,$$

where σ_t is a chosen hyperparameter and z is sampled from a standard Gaussian.

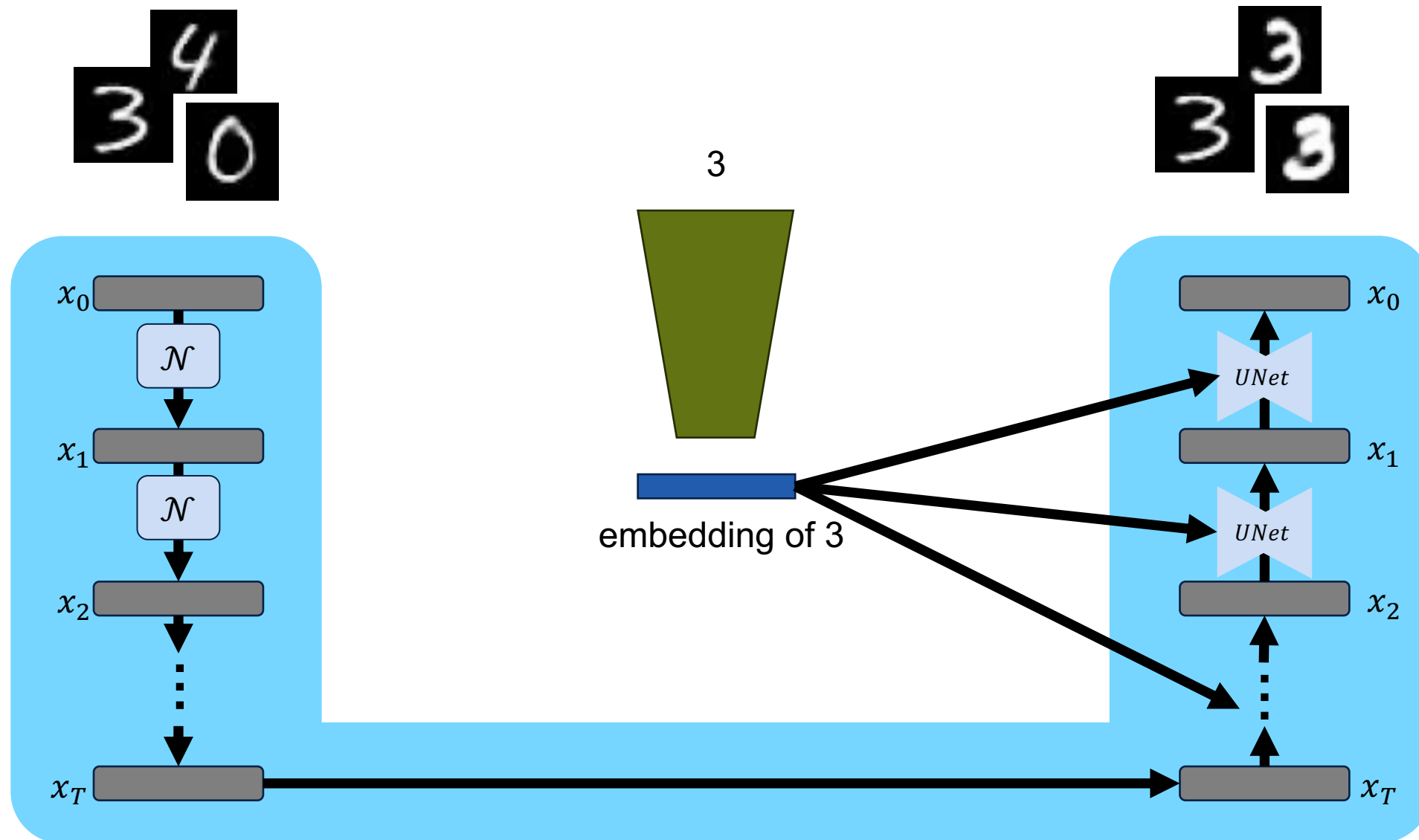


Conditional diffusion processes

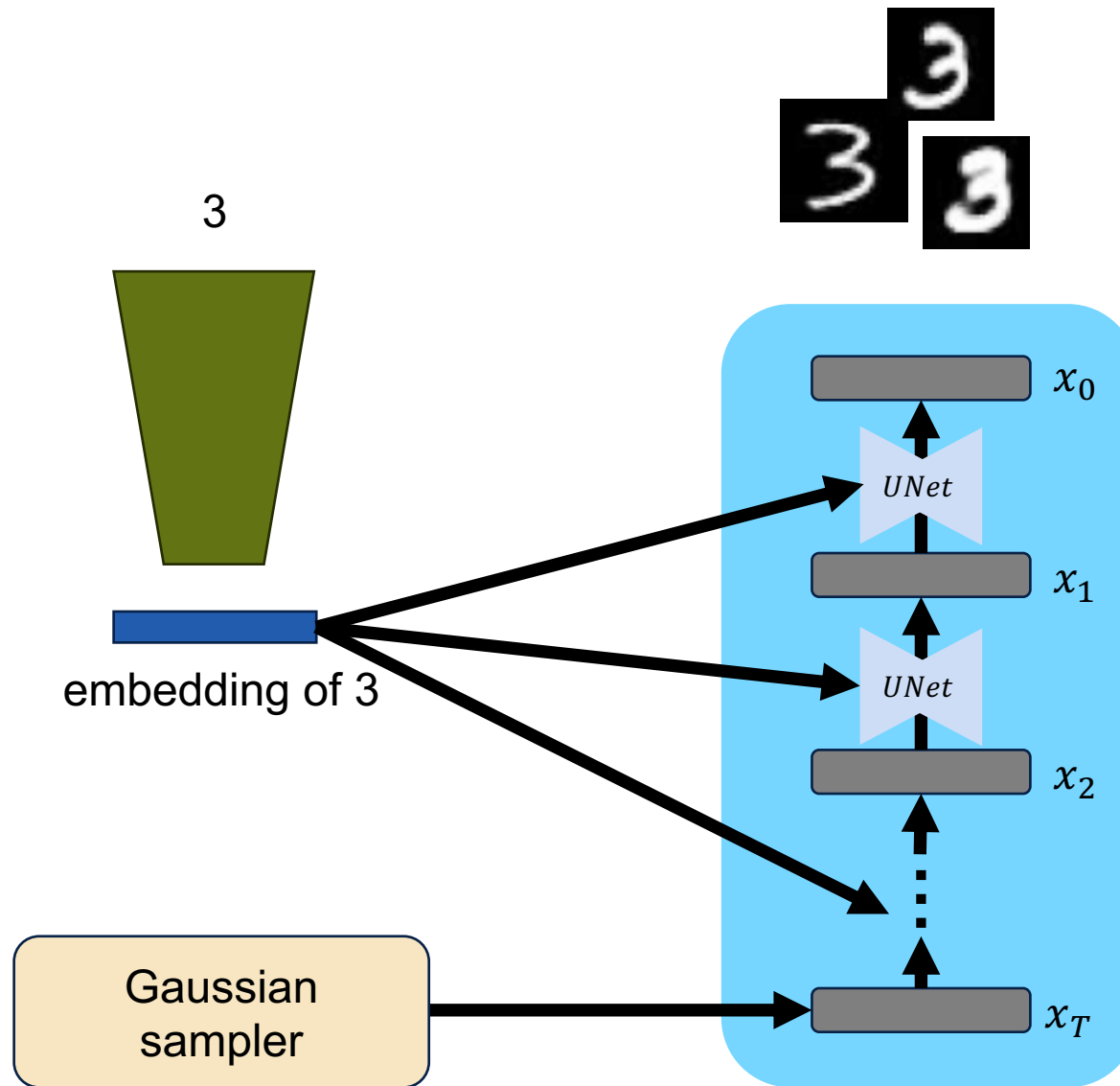
Diffusion process for MNIST



Conditional diffusion process for MNIST



Conditional generation of handwritten digits



Link to implementation with facehugger library

- <https://huggingface.co/learn/diffusion-course/en/unit2/3>